

Muhammad Raahim Rizwan

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Technical Skills

- **Languages:** Python, C++, SQL, JavaScript, HTML/CSS, Urdu (Native), English (Fluent).
- **Frameworks & Libraries:** Scikit-Learn, TensorFlow, FastAPI, Flask, Next.js, React, React Native.
- **Databases & DevOps:** MongoDB, Qdrant (Vector DB), Docker, Azure, Git.
- **Core Concepts:** Data Structures & Algorithms (DSA), Competitive Programming, MLOps, RAG Architecture.

Professional Experience

10Pearls Pakistan

Karachi, Pakistan

Data Science Intern — Project [GitHub](#)

May 2025 – June 2025

- Engineered an end-to-end MLOps pipeline for air quality index (AQI) forecasting using OpenWeather API, evaluating XGBoost, Random Forest, and Linear Regression to achieve an optimal 94% prediction accuracy and reduced RMSE by 6.4%.
- Engineered automated data preprocessing and feature transformation pipelines, significantly minimizing ingestion anomalies and stabilizing real-world environmental data streams for model consumption.
- Built and deployed a production-grade MongoDB-based model registry to systematically track experiments and manage model versioning.

NED University of Engineering and Technology

Karachi, Pakistan

Data Science Intern — Project [GitHub](#)

June 2025 – July 2025

- Architected a Retrieval-Augmented Generation (RAG) chatbot for university document automation, utilizing PyPDF2 for structured semantic parsing of complex academic content.
- Designed a high-performance semantic search pipeline combining `all-MiniLM-L6-v2` sentence embeddings and a Qdrant vector database.
- Delivered a full-stack production application featuring a lightweight Flask backend and an interactive React frontend to provide real-time, context-aware answers over institutional knowledge bases.

Leadership & Selected Projects

Lead Data Insight Club

CSIT Department, NED University

Student Director / Organizer

Dec 2023 – Present

- Orchestrated the department's inaugural Data Science Hackathon ("Data Quest"), managing budget allocation, corporate outreach, and engagement for over 100+ student participants.
- Spearheaded a series of technical workshops and knowledge-sharing sessions focused on practical machine learning applications, boosting departmental engagement with MLOps frameworks.

Matrimonial Platform Startup & Game Development

Personal Portfolio

Lead Full-Stack & AI Developer — [GitHub: Matrimonial App](#) · [NED Fighter](#)

Jan 2026 – Present

- Developing a professional-grade matrimonial platform leveraging Next.js and React Native to deliver a secure, highly responsive UI tailored for specialized career demographics.
- Architecting "NED Fighter," a 2D fighting game featuring custom sprite states and scene architecture built via the Phaser game engine, managing high-performance frame animation loops.

Education

NED University of Engineering and Technology

Karachi, Pakistan

BSCS, Data Science — **CGPA: 3.829**

Expected Graduation: October 2027

- **Relevant Frameworks & Coursework:** Advanced Data Structures, Design & Analysis of Algorithms, Object-Oriented Programming (C++/Python), Predictive Analytics.

Adamjee Govt. Science College

Karachi, Pakistan

Higher Secondary School Certificate (Pre-Engineering) — **Aggregate: 87.44%**

August 2023

Certifications

- **Google AI Professional Certification**
- **Data Structures and Algorithms Certification** – Simplilearn